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PROJECT REPORT

ESP32 Access Point & Raspberry Pi Two-Way Communication System

Project Overview

This document explains how to create a local communication channel between a Raspberry Pi and a laptop using a WiFi module or hotspot without internet access. The communication will use TCP socket programming and a specific port number so both devices can send and receive messages.

Objective

- ✓ Implement a Wi-Fi Access Point using ESP32
- ✓ Establish real-time two-way communication between Laptop and Raspberry Pi
- ✓ Demonstrate message sending, receiving, and auto-reply (optional)
- ✓ Prepare for project demo / viva.

Components Required:

Hardware Required

- ✓ ESP32 – Acts as Wi-Fi Access Point for IoT communication.
- ✓ Raspberry Pi – Server / Client to send and receive messages.
- ✓ Laptop – Client to send messages and test communication.
- ✓ USB Cables / Power Supply – To power ESP32 and Raspberry Pi.

Software Required

- ✓ Arduino IDE – To program the ESP32.
- ✓ Python 3.x – To write Raspberry Pi server and laptop client code.
- ✓ VS Code / Notepad – For editing code files.
- ✓ Serial Monitor – To debug ESP32 and check WiFi IP.

SERVER CODE:

```
#include <WiFi.h>

const char* ssid = "Network";
const char* password = "12345678";

void setup() {
Serial.begin(115200);

WiFi.softAP(ssid, password);

Serial.println("ESP32 Access Point Started");
Serial.print("IP Address: ");
Serial.println(WiFi.softAPIP());
}

void loop() {
}
```

CLIENT CODE:

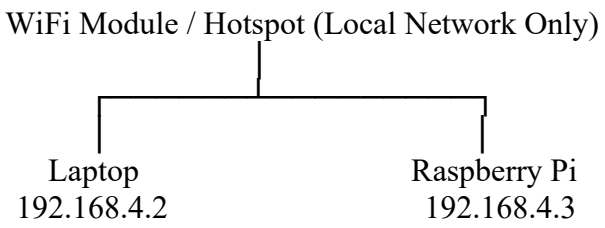
```
import socket
import threading

host = "192.168.4.3"   # Raspberry Pi IP
port = 5000

client = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
client.connect((host, port))

print("Connected to Raspberry Pi")
```

```
def receive():
    while True:
        data = client.recv(1024).decode()




        if data.lower() == "exit":
            break
        print("Raspberry Pi:", data)

def send():
    while True:
        msg = input("You: ")
        client.send(msg.encode())
        if msg.lower() == "exit":
            break

threading.Thread(target=receive).start()
threading.Thread(target=send).start()
```

ESP32 CODE:

```
#include <WiFi.h>

const char* ssid = "ESP32_Network";
const char* password = "12345678";

void setup() {
  Serial.begin(115200);

  WiFi.softAP(ssid, password);

  Serial.println("ESP32 Access Point Started");
  Serial.print("IP Address: ");
  Serial.println(WiFi.softAPIP());
}

void loop() {
}
```

NETWORK ARCHITECTURE :



Both devices are connected to the same local network and communicate using a port.

Step 1 : Setup Devices

- ✓ Turn ON the Raspberry Pi and Laptop.
- ✓ Create a WiFi hotspot or use a WiFi module.
- ✓ Connect both Raspberry Pi and Laptop to the same WiFi network.

Step 2: Find Raspberry Pi IP Address

- ✓ Open the terminal in Raspberry Pi.
- ✓ Type the command:

  hostname -I
- ✓ Note the IP address (example: 192.168.4.3).

Step 3: Create Server Program on Raspberry Pi

- ✓ Open a new Python file named server.py.
- ✓ Write the socket server code.
- ✓ Run the program using:

  python3 server.py
- ✓ The Raspberry Pi will wait for connection.

Step 4: Create Client Program on Laptop

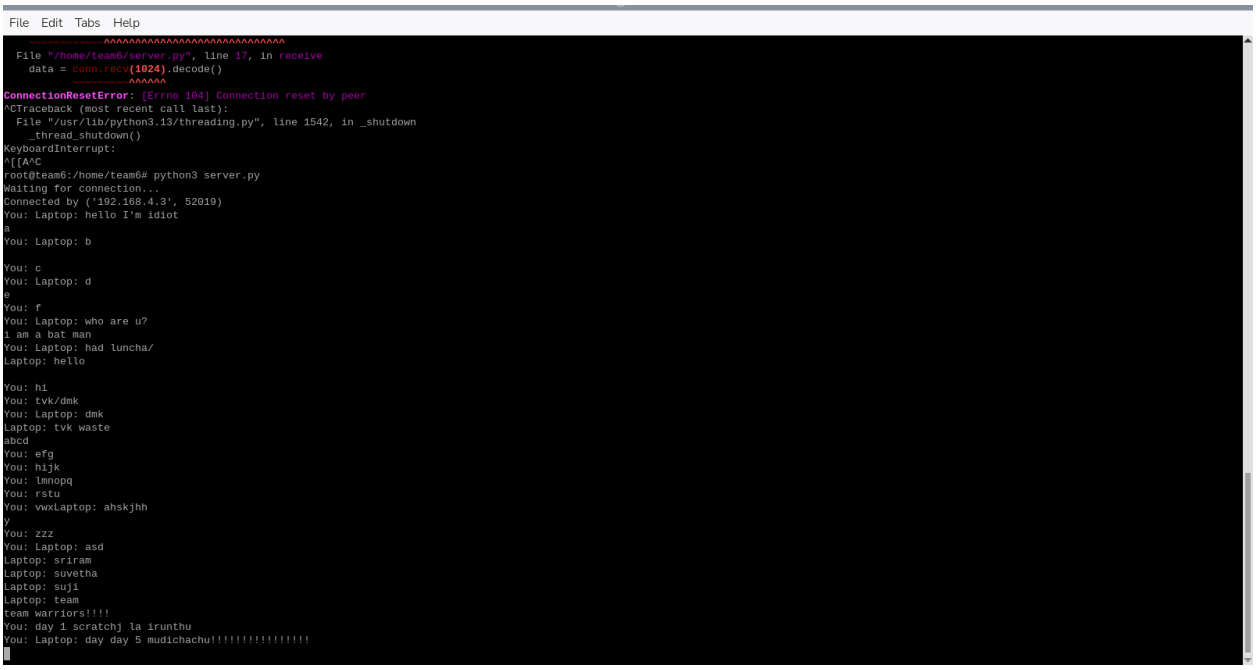
- ✓ Open a Python file named client.py.
- ✓ Write the socket client code.
- ✓ Use the Raspberry Pi IP address in the code.
- ✓ Run the program using:

python3 client.py

Step 5: Communication Between Devices

- ✓ The laptop sends a message to Raspberry Pi.
- ✓ Raspberry Pi receives and displays the message.
- ✓ Communication happens through TCP socket using a specific port number (5000).

Expected Output:



```
File  Edit  Tabs  Help
    ~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/team6/server.py", line 17, in receive
    data = conn.recv(1024).decode()
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
ConnectionResetError: [Errno 104] Connection reset by peer
^CTraceback (most recent call last):
  File "/usr/lib/python3.13/threading.py", line 1542, in _shutdown
    _thread_shutdown()
KeyboardInterrupt:
^[[A^C
root@team6:/home/team6# python3 server.py
Waiting for connection...
Connected by ('192.168.4.3', 52019)
You: Laptop: hello I'm idiot
a
You: Laptop: b

You: c
You: Laptop: d
e
You: f
You: Laptop: who are u?
i am a bat man
You: Laptop: had luncha/
Laptop: hello

You: hi
You: tvk/dmk
You: Laptop: dmk
Laptop: tvk waste
abcd
You: efg
You: hijk
You: lmnopq
You: rstu
You: vwxLaptop: ahskjhh
y
You: zzz
You: Laptop: asd
Laptop: sriram
Laptop: suvetha
Laptop: suji
Laptop: team
team warriors!!!!
You: day 1 scratchj la irunthu
You: Laptop: day day 5 mudichachu!!!!!!!!!!!!!!!
```

```
KeyboardInterrupt:
PS C:\ras> python3 client.py
Connected to Raspberry Pi
You: hello I'm idiot
You: Raspberry Pi: a
b
You: Raspberry Pi: c
d
You: Raspberry Pi: e
who are u?Pi: f
You: Raspberry Pi: i am a bat man
had luncha/
You: hello
You:    aspberry Pi: hi
You: Raspberry Pi: tvk/dmk
dmk
You: tvk waste
You: Raspberry Pi: lmnopq
You: erry Pi: efg
You: ahskjhherry Pi: rstu
You: Raspberry Pi: vwxy
Raspberry Pi: zzz
asd
You: sriram
You: suvetha
You: suji
You: team
You: Raspberry Pi: team warriors!!!!
day Raspberry Pi: day 1 scratchj la irunthu
    day 5 mudichachu!!!!!!!!!!!!!!!
You:
```

Result:

The ESP32 successfully created a WiFi Access Point, and both the Raspberry Pi and laptop connected to the network. TCP socket communication was established, allowing real-time two-way message exchange between the laptop and Raspberry Pi through the specified port.